

$$\begin{aligned}
C_{\text{HW}} \rightarrow & \frac{1}{16\pi^2} \left(-\frac{1}{16} g_2^2 \left(2 c_\gamma^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + \sum_{\mathbf{p}} c_{2\gamma} g_1^2 \right) \text{LF}_{3,0}[\mathbf{m}_l^{\text{p}}] + \frac{1}{16} g_2^2 \left(2 c_\gamma^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + \sum_{\mathbf{p}} c_{2\gamma} g_1^2 \right) \right. \\
& \text{LF}_{4,-1}[\mathbf{m}_l^{\text{p}}] + \frac{1}{16} g_2^2 \left(-6 c_\gamma^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} - 6 s_\gamma^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} + \sum_{\mathbf{p}} c_{2\gamma} g_1^2 \right) \text{LF}_{3,0}[\mathbf{m}_q^{\text{p}}] + \\
& \frac{1}{16} g_2^2 \left(6 c_\gamma^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} + 6 s_\gamma^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} - \sum_{\mathbf{p}} c_{2\gamma} g_1^2 \right) \text{LF}_{4,-1}[\mathbf{m}_q^{\text{p}}] + \\
& \frac{1}{64} \left(g_1^2 g_2^2 (1 + 3 c_{4\gamma}) + 3 g_2^4 (-1 + c_{4\gamma}) \right) \text{LF}_{3,0}[\mathbf{m}_\pm] - \\
& \frac{1}{64} g_2^2 \left(g_1^2 (1 + 3 c_{4\gamma}) + 3 g_2^2 (-1 + c_{4\gamma}) \right) \text{LF}_{4,-1}[\mathbf{m}_\pm] + \frac{5}{4} g_2^4 \text{LF}_{2,2,-1}[\mathbf{m}_2, \tilde{\mu}] + \\
& \mathbf{m}_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{2,2,0}[\mathbf{m}_2, \tilde{\mu}] + g_2^4 \text{LF}_{3,1,-1}[\mathbf{m}_2, \tilde{\mu}] - g_2^4 \text{LF}_{3,2,-2}[\mathbf{m}_2, \tilde{\mu}] - \\
& 2 \mathbf{m}_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{3,2,-1}[\mathbf{m}_2, \tilde{\mu}] - g_2^4 \text{LF}_{4,1,-2}[\mathbf{m}_2, \tilde{\mu}] - 2 \mathbf{m}_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{4,1,-1}[\mathbf{m}_2, \tilde{\mu}] - \\
& \frac{1}{4} g_2^2 \left(c_\gamma \overline{a_e^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_e^{\text{pr}}} \right) \left(c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}} \right) \text{LF}_{4,1,-1}[\mathbf{m}_l^{\text{p}}, \mathbf{m}_e^{\text{r}}] + \\
& \frac{1}{4} g_2^2 \left(c_\gamma \overline{a_e^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_e^{\text{pr}}} \right) \left(c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}} \right) \text{LF}_{5,1,-2}[\mathbf{m}_l^{\text{p}}, \mathbf{m}_e^{\text{r}}] - \\
& \frac{3}{4} g_2^2 \left(c_\gamma \overline{a_d^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_d^{\text{pr}}} \right) \left(c_\gamma a_d^{\text{pr}} - s_\gamma \tilde{\mu} y_d^{\text{pr}} \right) \text{LF}_{4,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] + \\
& \frac{3}{4} g_2^2 \left(c_\gamma \overline{a_d^{\text{pr}}} - s_\gamma \tilde{\mu} \overline{y_d^{\text{pr}}} \right) \left(c_\gamma a_d^{\text{pr}} - s_\gamma \tilde{\mu} y_d^{\text{pr}} \right) \text{LF}_{5,1,-2}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] - \\
& \frac{3}{8} g_2^2 \left(s_\gamma \overline{a_u^{\text{pr}}} - \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}} \right) \left(s_\gamma a_u^{\text{pr}} - \tilde{\mu} c_\gamma y_u^{\text{pr}} \right) \text{LF}_{2,2,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] - \\
& \frac{3}{8} g_2^2 \left(s_\gamma \overline{a_u^{\text{pr}}} - \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}} \right) \left(s_\gamma a_u^{\text{pr}} - \tilde{\mu} c_\gamma y_u^{\text{pr}} \right) \text{LF}_{3,1,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] + \\
& \frac{3}{8} g_2^2 \left(s_\gamma \overline{a_u^{\text{pr}}} - \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}} \right) \left(s_\gamma a_u^{\text{pr}} - \tilde{\mu} c_\gamma y_u^{\text{pr}} \right) \text{LF}_{3,2,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] + \\
& \frac{3}{8} g_2^2 \left(s_\gamma \overline{a_u^{\text{pr}}} - \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}} \right) \left(s_\gamma a_u^{\text{pr}} - \tilde{\mu} c_\gamma y_u^{\text{pr}} \right) \text{LF}_{4,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] + \\
& \frac{3}{8} g_2^2 \left(s_\gamma \overline{a_u^{\text{pr}}} - \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}} \right) \left(s_\gamma a_u^{\text{pr}} - \tilde{\mu} c_\gamma y_u^{\text{pr}} \right) \text{LF}_{3,1,0}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \\
& \frac{3}{8} g_2^2 \left(s_\gamma \overline{a_u^{\text{pr}}} - \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}} \right) \left(s_\gamma a_u^{\text{pr}} - \tilde{\mu} c_\gamma y_u^{\text{pr}} \right) \text{LF}_{3,2,-1}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \\
& \frac{9}{8} g_2^2 \left(s_\gamma \overline{a_u^{\text{pr}}} - \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}} \right) \left(s_\gamma a_u^{\text{pr}} - \tilde{\mu} c_\gamma y_u^{\text{pr}} \right) \text{LF}_{4,1,-1}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \\
& \frac{3}{4} g_2^2 \left(s_\gamma \overline{a_u^{\text{pr}}} - \tilde{\mu} c_\gamma \overline{y_u^{\text{pr}}} \right) \left(s_\gamma a_u^{\text{pr}} - \tilde{\mu} c_\gamma y_u^{\text{pr}} \right) \text{LF}_{5,1,-2}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \\
& \frac{1}{8} g_1^2 g_2^2 \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_1] - \frac{1}{4} \mathbf{m}_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 g_2^2 \text{LF}_{3,1,0}[\tilde{\mu}, \mathbf{m}_1] + \frac{3}{8} g_1^2 g_2^2 \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_1] + \\
& \frac{1}{2} \mathbf{m}_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 g_2^2 \text{LF}_{4,1,-1}[\tilde{\mu}, \mathbf{m}_1] - \frac{1}{4} g_1^2 g_2^2 \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_1] - \\
& \frac{1}{2} \mathbf{m}_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 g_2^2 \text{LF}_{5,1,-2}[\tilde{\mu}, \mathbf{m}_1] - \frac{3}{8} g_2^4 \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_2] - \frac{3}{4} \mathbf{m}_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{3,1,0}[\tilde{\mu}, \mathbf{m}_2] - \\
& g_2^4 \text{LF}_{3,2,-2}[\tilde{\mu}, \mathbf{m}_2] - 2 \mathbf{m}_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_2] + \frac{9}{8} g_2^4 \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_2] + \\
& \left. \frac{3}{2} \mathbf{m}_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{4,1,-1}[\tilde{\mu}, \mathbf{m}_2] - \frac{3}{4} g_2^4 \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_2] - \frac{3}{2} \mathbf{m}_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{5,1,-2}[\tilde{\mu}, \mathbf{m}_2] \right)
\end{aligned}$$